Jeet Bahadur Thapa
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Dayton, Ohio | 937-414-7125 (Open to Relocate)

EDUCATION:

- **Master of Science in Mechanical Engineering** — Jan 2024 – Present
  Wright State University, Dayton, Ohio — Cumulative GPA: 3.857 / 4.0

SOFTWARE AND TECHNICAL SKILLS:

- *CAD/Simulation*: SolidWorks, Simufact, ANSYS, MATLAB, Python
- *Testing Equipment*: Formlabs Fuse 1 (SLS printer), FDM printer, (UTS tensile test), Tensile test (UTS), MTS (ASTM D695 compression test), Keyence VR6300 (roughness), Mitutoyo Vision System (dimension), SEM, XRD
- *Vibration Analysis:* Spectral Dynamics Bobcat analyzer, RMS/time-domain/frequency-domain analysis

CERTIFICATIONS, AWARDS & RESEARCH PRESENTATIONS:

- CSWA, CSWP, CSWA-AM, CSWPA-DT, Oasis Rapid Certification on Semiconductor & Integrated manufacturing
- Conferences for Poster Presenter for 3D printing (SLS) and Polymer characterization:
  - MS&T 2025, DESS 2025, PINO 2026

PROFESSIONAL EXPERIENCE:

Graduate Research Assistant: Wright State University, Dayton, Ohio — May 2025 – *Present*

Additive Manufacturing with Mechanical and Physical Characterization

- Ran ASTM D695 compression on SLS-printed PA12/PA12-GF at 0.01–5 s^{-1} (MTS 244.51) and conducted ISO 179-1 Charpy impact (Instron Ceast 9050) to quantify modulus, yield, stress–strain response, and toughness across build orientations. Performed metrology and surface characterization using Mitutoyo Vision System and Keyence VR-6000 (Sa/Sq roughness/topography).SLS Printing
- Completed XRD on PA12 powder vs printed parts for crystalline phase comparison and SEM fractography on PA12-GF after cryogenic fracture to assess fracture morphology and glass bead microstructure.

Graduate Teaching Assistant (GTA) Wright State University, Dayton, Ohio

Mechanical Vibrations Lab: — Aug 2025 – Dec 2025

Performed DSC to extract Tm, Tc, and ΔHm for melting/crystallization behavior (baseline thermal transitions/crystallinity) and TGA to generate mass-loss/degradation onset curves benchmarking thermal stability for PA12 vs PA12-GF.

Mechanical and Material Testing Lab: — Jan 2025 – April 2025

- Tensile Test (Instron), Elastic Beam Bending, Heat Treatment Effect, Impact Test (Charpy), Fatigue Test

Technical Advisor & Sr. Senior Engineer: B & K Enterprises, Bhaktapur, Nepal — Dec 2018 – Nov 2023

- Implemented standardized vehicle service SOPs across multiple vehicle classes, improving shop throughput and saving 225+ labor hours/year; led component failure analysis/RCA (hydraulics, clutch, brake systems) and preventive actions reducing repeat failures by ~15%.
- Delivered technical operations support through warranty/inventory reporting (Microsoft Office, Microsoft Dynamics Navision), team coordination for cost/quality targets, and emissions compliance inspections using exhaust gas analyzers (gasoline) and opacity meters (diesel

Service Advisor: Sipradi Trading Pvt. Ltd. Kathmandu, Nepal — Jul 2016 – Dec 2018

- Improved service turnover by ~20% by streamlining diagnostics using TATA Diagnostics Software (DTCs, live data, actuator tests, wiring checks) and implementing structured job cards/repair planning. Deployed Hunter Hawkeye Elite 3D alignment to improve camber/caster/toe accuracy with post-service verification, and trained technicians on updated OEM procedures after TATA Motors HQ programs.

PROJECTS:

- Retrofitting of a 2-stroke Gasoline Engine to a Compressed Air Engine — Nov 2014 – Nov 2015
- 3D Analysis of the AGARD 445.6 Wing by FEA for Vibration & Flutter Prediction — Oct 2024- Nov 2024
- Simulation of Metal Forming to Analyze the Effects of Geometry and Friction Using Simufact